

AFL Assistant Executive Report

1. Product Goal

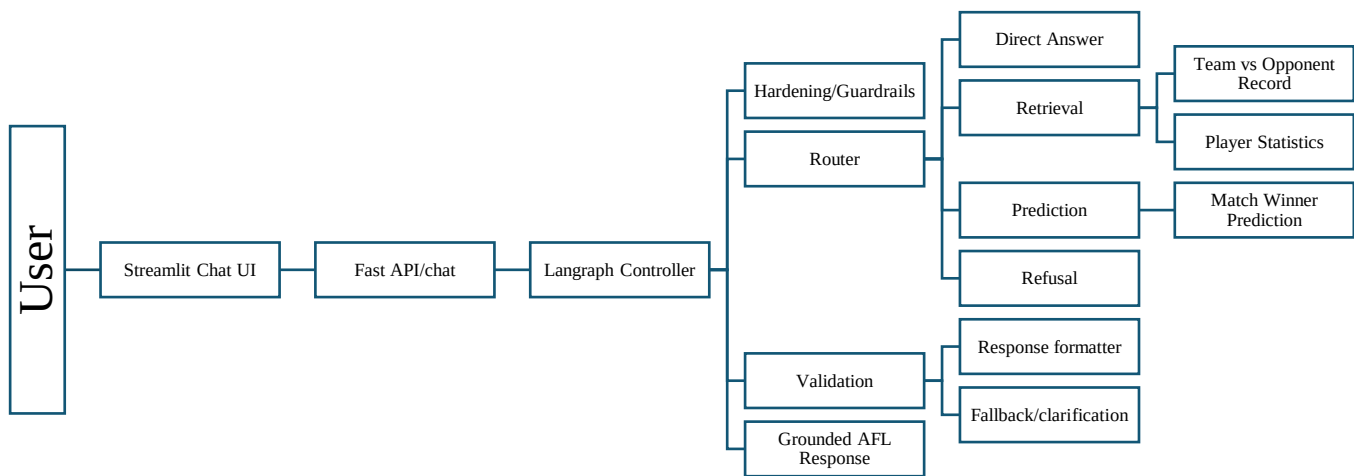
The AFL Assistant is a domain-scoped AI assistant designed to answer questions about Australian Football League (AFL) teams, players, matches, statistics, history, rules, and match predictions.

The main objective is to combine conversational AI with reliable structured AFL data rather than relying on the language model to generate numerical statistics from memory. The system therefore uses deterministic retrieval tools for factual questions and a trained machine-learning model for match-winner predictions.

The assistant is also designed to remain within the AFL domain. Off-topic requests and prompt-injection attempts are rejected through explicit guardrails.

2. System Architecture

The final system uses **LangGraph as the orchestration layer**, with FastAPI providing the backend API and Streamlit providing the user-facing interface.



Chat and Retrieval

For factual AFL questions, the assistant routes requests to structured Pandas-based retrieval tools. This approach is appropriate for numerical AFL statistics because the data is structured and exact values can be retrieved directly from the dataset.

For example:

“What is Geelong’s record against Essendon?”

The retrieval tool returns the underlying statistics, producing:

39 wins, 23 losses, and 1 draw across 63 matches.

Conversation history is retained using a conversation ID, allowing follow-up questions such as:

“How many wins did they have?”

to be interpreted using the previous conversation context.

Prediction

Prediction requests are routed to the match-winner prediction tool. The tool resolves the teams and fixture, obtains the appropriate pre-match features, and calls the Day 2 prediction model.

The prediction response includes the predicted winner, probability, match date, teams, and class probabilities.

A prediction disclaimer is included because model probability represents a statistical prediction rather than certainty.

API and Interface

FastAPI exposes the assistant through a /chat endpoint with Pydantic request/response validation. The API also records structured operational information including route, intent, tools called, latency, and errors.

The Streamlit interface provides a simple conversational frontend with:

- AFL chat interaction
- Conversation memory
- Prediction metadata
- Route/intent/latency information
- Quick example questions
- New-conversation controls

3. Evaluation Results

The assistant was evaluated using **30 test cases** covering factual AFL questions, prediction sanity checks, scope guardrails, and multi-turn conversations.

Evaluation Category	Tests	Passed	Pass Rate
Factual Q&A	8	4	50.0%
Prediction Sanity	7	7	100%
Scope Guardrails	7	7	100%
Multi-turn Coherence	8	4	50.0%
Overall	30	22	73.3%

The prediction system was additionally evaluated on a **2025 holdout set of 216 matches**.

- Day 2 model accuracy: **67.13%**
- Same-match ladder baseline: **65.28%**
- Model improvement over baseline: **+1.85 percentage points**
- Successful predictions: **216/216**
- Prediction errors during evaluation: **0 tool/runtime errors**

A separate full-dataset ladder-based baseline achieved **63.08% accuracy**.

The evaluation shows that the prediction pipeline is operational and provides measurable predictive performance, while the conversational layer still has areas requiring improvement.

4. Known Limitations

Data Recency

The assistant is dependent on the available AFL datasets. New match results and current-season information must be added to the feature tables before the system can provide up-to-date predictions or statistics.

Model Accuracy Ceiling

The prediction model does not guarantee match outcomes. The 2025 holdout accuracy of 67.13% demonstrates useful predictive signal, but approximately one-third of evaluated predictions were incorrect. AFL matches also contain uncertainty that cannot be completely captured by the available features.

Retrieval and Intent Mapping

The evaluation identified failures where different natural-language forms of the same AFL question were not always mapped to the correct retrieval tool.

For example, questions involving wins, losses, or draws can require better query interpretation.

Multi-turn Context

Multi-turn evaluation achieved 50% accuracy. Some follow-up questions were not correctly connected to the previous question, particularly follow-ups involving prediction probability, prediction certainty, and match-record details.

Player Statistics

Some player-statistics requests require a season/year. When the year is missing, the system may need to ask the user for clarification instead of attempting the tool call.

Guardrail Edge Cases

The system contains explicit off-topic and prompt-injection checks, but rule-based guardrails cannot guarantee detection of every possible adversarial phrasing. Continued adversarial evaluation is therefore required.

5. Recommended Next Steps

1. Improve Natural-Language Retrieval Mapping

Expand the retrieval layer so that equivalent AFL questions map to the same structured tool. Examples include:

- “How many times did the Cats beat the Bombers?”
- “How many wins does Geelong have against Essendon?”
- “What about their losses?”
- “How many draws were there?”

2. Improve Conversation Context Resolution

Store structured context such as:

- Teams previously discussed
- Match being predicted
- Prediction probability
- Current topic
- Requested statistic

This would improve follow-up questions without requiring the user to repeat the full context.

3. Refresh AFL Data Regularly

Implement the weekly refresh process so completed AFL rounds automatically feed into the feature table and prediction evaluation pipeline.

4. Monitor Model Drift

Evaluate prediction accuracy after each completed round and retrain when enough new data is available or sustained performance degradation is observed.

5. Expand Evaluation

Increase the evaluation suite with more adversarial prompts, multi-turn conversations, ambiguous team names, missing dates, missing seasons, and newly introduced AFL data.

6. Production Deployment

Move the current Colab-based demonstration deployment to a persistent hosting environment for continuous availability. FastAPI and Streamlit can then be deployed as production services rather than relying on a temporary notebook runtime.

Conclusion

The AFL Assistant demonstrates an end-to-end domain-specific AI system combining **LangGraph orchestration, structured AFL retrieval, machine-learning prediction, guardrails, FastAPI, conversation memory, monitoring, and a Streamlit interface.**

The current system achieved an overall evaluation pass rate of **73.3%**, with **100% of prediction sanity tests and 100% of scope-guardrail tests passing.** The main improvement areas are natural-language retrieval mapping and multi-turn context resolution.

The project provides a working foundation that can be further improved through better context management, regular data updates, continued evaluation, and persistent deployment.